

Shape-Informed Clustering of Multi-Dimensional Functional Data via Deep Functional Autoencoders

Samuel V. Singh¹, Shirley Coyle², Mimi Zhang¹

¹School of Computer Science and Statistics, Trinity College Dublin

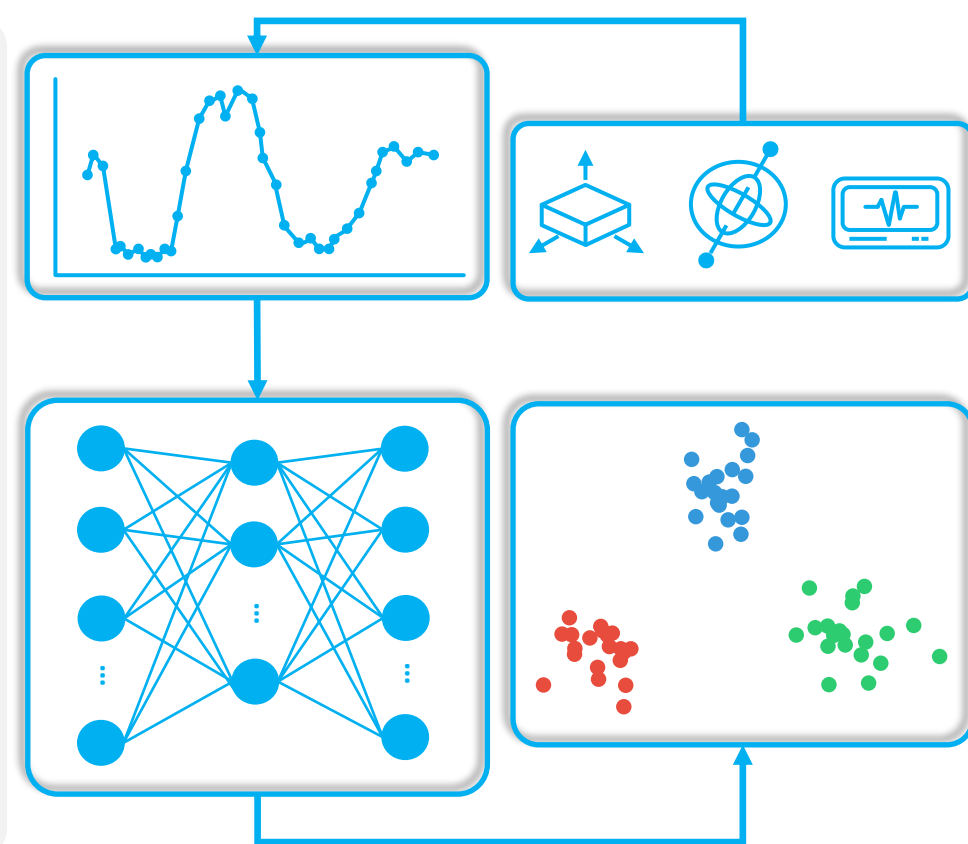
²School of Electronic Engineering, Dublin City University

Introduction

Multi-dimensional functional data (FD) are vector-valued functions observed over a continuous domain (e.g., time or space). Existing multi-dimensional FD clustering methods typically either compress the data via functional PCA or construct pairwise (dis)similarity measures and then apply a multivariate clustering algorithm; however, these approaches often struggle when dimensionality is high, cross-component dependencies are nonlinear, or component functions are misaligned (phase variation).

FAEclust:

- (i) encoder and decoder are both universal approximators;
- (ii) the shape-informed clustering objective is robust to phase variation
- (iii) the shape of the functional weights reveals which regions of the input functional data contribute most.



Function → Vector → Function

Encoder $\mathcal{E} : \mathcal{H}(\mathcal{T}, \mathbb{R}^p) \rightarrow \mathbb{R}^s$

$$\mathbf{x}^{(1)} = \mathbf{a} \left(\int_{\mathcal{T}} \mathbf{W}^{(1)}(t) \mathbf{y}(t) dt + \mathbf{b} \right)$$

$$\mathbf{x} = \text{MLP}(\mathbf{x}^{(1)})$$

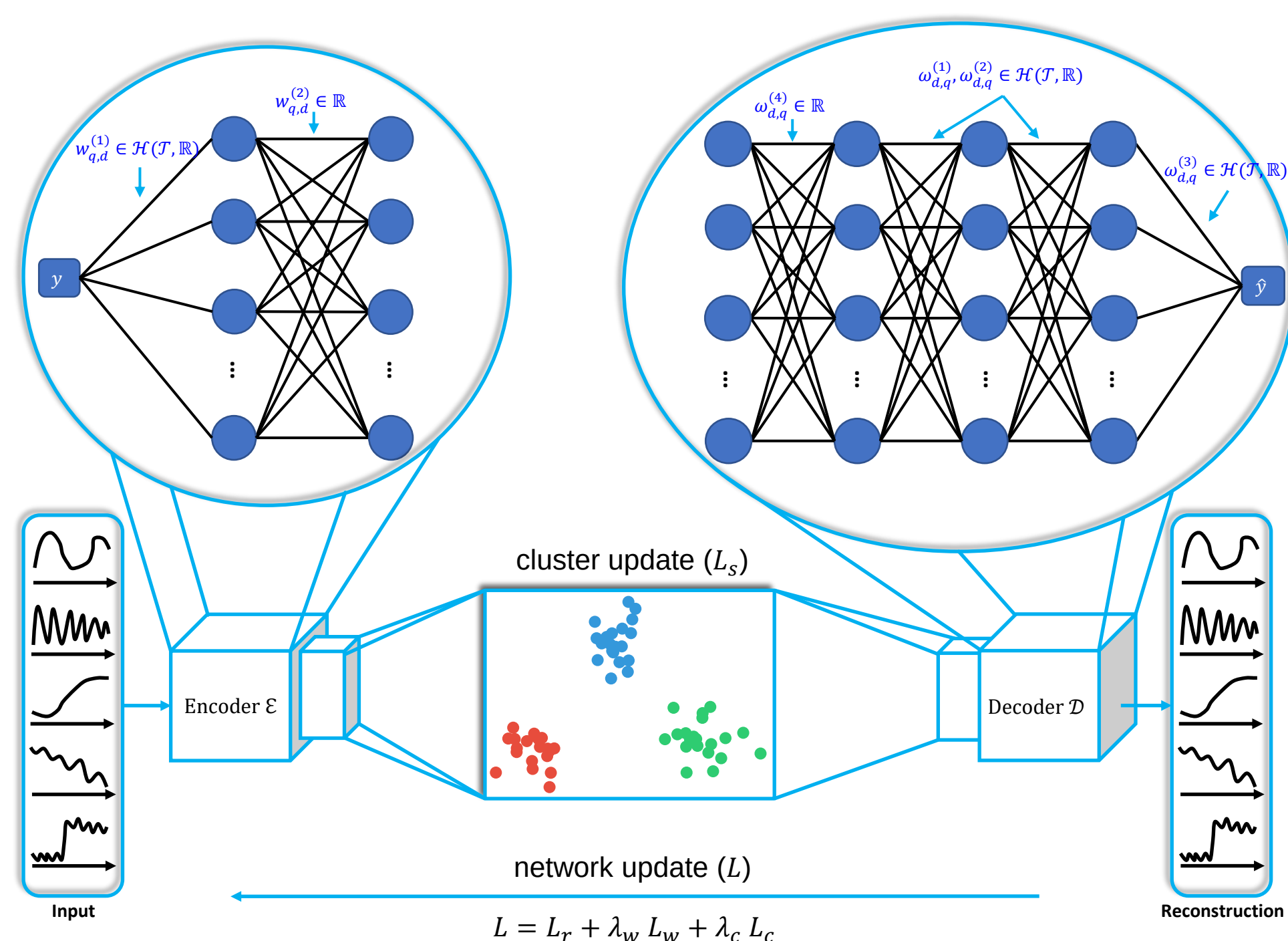
Decoder $\mathcal{D} : \mathbb{R}^s \rightarrow \mathcal{H}(\mathcal{T}, \mathbb{R}^p)$

$$\hat{\mathbf{x}}^{(1)} = \text{MLP}(\mathbf{x})$$

$$\hat{\mathbf{y}}^{(1)}(t) = \mathbf{a}(\mathbf{W}^{(1)}(t) \hat{\mathbf{x}}^{(1)} + \mathbf{b}_1(t))$$

$$\hat{\mathbf{y}}^{(2)}(t) = \mathbf{a}(\mathbf{W}^{(2)}(t) \hat{\mathbf{y}}^{(1)}(t) + \mathbf{b}_2(t))$$

$$\hat{\mathbf{y}}(t) = \mathbf{W}^{(3)}(t) \hat{\mathbf{y}}^{(2)}(t)$$



Joint network training and clustering

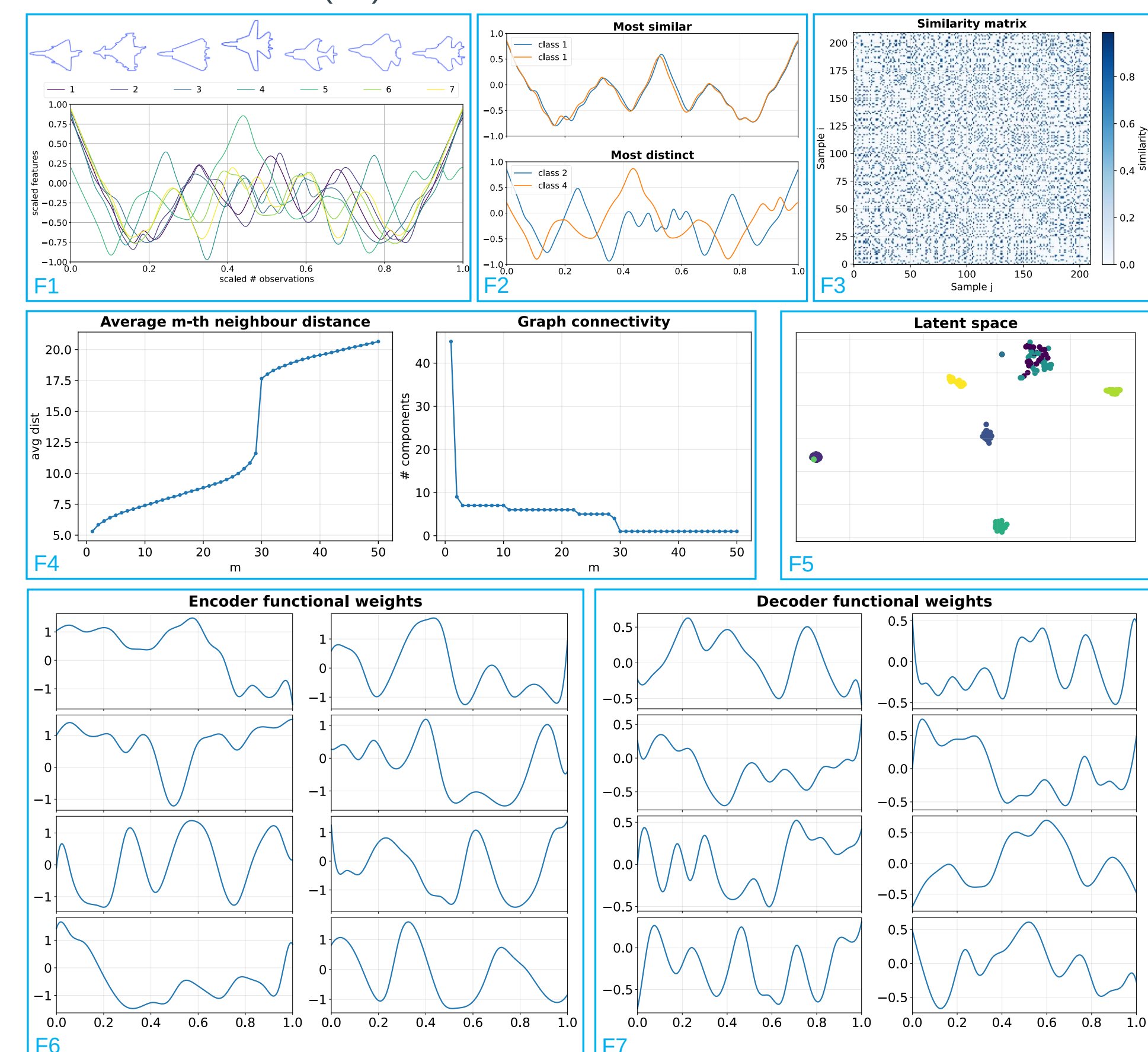
- 1. Forward (cluster update):** update clusters via convex clustering $L_s = \frac{1}{n} \|\mathbf{X} - \mathbf{U}\|_F^2 + \lambda \sum_{i < j} s(\mathbf{y}_i, \mathbf{y}_j) \|\mathbf{u}_i - \mathbf{u}_j\|_{\ell_1}$, where the similarity measure $s(\mathbf{y}_i, \mathbf{y}_j)$ depends on the elastic distance between $\mathbf{y}_i$ and $\mathbf{y}_j$.
 - The path-following homotopy algorithm obtains a hierarchy of agglomerative clustering solutions in $O(n \log(n))$ time.
 - From the hierarchy, we choose the optimal clustering using an internal validation index (e.g., silhouette or Davies-Bouldin).
- 2. Backward (network update):** minimize an integrated objective function $L = L_r + \lambda_w L_w + \lambda_c L_c$, where L_r is the reconstruction loss, L_w enforces smooth or orthogonal functional weights, L_c encourages within-cluster compactness and between-cluster separation.
- 3. Iteration:** alternate cluster updates and network updates -- pretrain with $(L_r + \lambda_w L_w)$, then fine-tune with full objective L .

Experiments

- (i) **Benchmarking methods:** FNN (Functional Neural Net), FAE (Functional AutoEncoder), VANO (Variational Autoencoding Neural Operator), funHDDC, funclust, FADPclust (FADP1/FADP2).
- (ii) **Euclidean FD:** 17 UEA/UCR datasets -- 9 one-dimensional and 8 multi-dimensional.
- (iii) **Manifold-valued FD:** simulated functions taking values on the hypersphere, hyperbolic space, and Swiss-roll manifold, plus trajectories from the Lorenz attractor and a pendulum.
- (iv) **Time-warped FD:** 12 simulated scenarios with random time warping, each replicated 100 times.

We use the Plane dataset from the UEA/UCR Time Series Archive to partition aircraft outlines into different groups. After rescaling, each outline is smoothed with B-splines (F1). F2 shows the most similar and most dissimilar trajectories under the elastic distance.

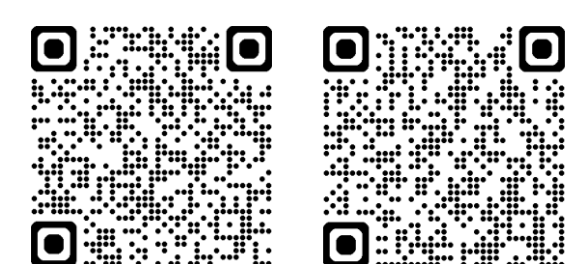
Pairwise elastic distances are converted into an m-nearest-neighbor (m-NN) graph, with m chosen automatically at the knee of the average m-th neighbor distance curve and at the point where graph connectivity stabilizes (F4). This graph defines the similarity matrix (F3). We then train FAEclust on the smoothed outlines to learn a structured latent space and select the optimal cluster partition from the hierarchy (F5). Finally, the learned functional weights in the decoder highlight which regions of the input functions drive the clustering (F6), and the final decoder layer reconstructs the aircraft outlines (F7).



Conclusion

FAEclust unifies deep functional representation learning with shape-aware convex clustering to deliver robust, accurate clustering of complex multi-dimensional FD. FAEclust consistently performs on Euclidean and manifold-valued FD and remains stable under substantial phase variation.

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Acknowledgement

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Datasets	funHDDC	fundust	FADP1	FADP2	FNN	FAE	VANO	FAEdust
BirdChicken	0.055	0.080	0.019	0.055	0.290	0.259	0.302	0.339
CBF	0.362	0.066	0.435	0.318	0.363	0.370	0.743	0.724
Chinatown	0.147	0.147	0.246	0.055	0.236	0.110	0.339	0.343
DSR	0.786	0.017	0.692	0.876	0.679	0.645	0.713	0.887
ECG200	0.173	0.101	0.201	0.143	0.214	0.213	0.217	0.261
Fungi	0.773	0.186	0.477	0.501	0.244	0.624	0.798	0.925
Plane	0.819	0.013	0.725	0.741	0.841	0.825	0.846	0.907
Rock	0.373	0.228	0.216	0.335	0.184	0.089	0.355	0.447
Symbols	0.767	0.001	0.435	0.712	0.748	0.800	0.817	0.824
Blink	0.410	0.048	0.189	0.177	0.453	0.506	0.522	0.633
BM	0.377	0.031	0.191	0.422	0.401	0.676	0.592	0.539
EOS	0.209	0.018	0.104	0.102	0.152	0.206	0.180	0.266
Epilepsy	0.143	0.028	0.077	0.225	0.274	0.209	0.297	0.485
ERing	0.735	0.012	0.288	0.714	0.643	0.743	0.733	0.664
FM	0.001	0.001	0.002	0.002	0.174	0.138	0.174	0.228
JV	0.840	0.069	0.294	0.466	0.236	0.854	0.899	0.893
SWJ	0.268	0.040	0.248	0.046	0.174	0.170	0.344	0.324
Hypersphere	0.016	0.478	0.137	0.067	0.089	0.307	0.443	0.737
Hyperbolic	0.005	0.013	0.001	0.001	0.004	0.047	0.410	0.798
Swiss roll	0.127	0.114	0.382	0.189	0.016	0.125	0.242	0.432
Lorenz	0.109	0.389	0.092	0.144	0.023	0.246	0.251	0.457
Pendulum	0.887	0.376	0.797	0.808	0.253	0.794	0.905	0.986